Northlight Weather API v1

Northlight serves current conditions, short-range forecasts and observation history for a network of weather stations along the Nordic coast. This handbook is one set of Markdown documents that cudoc publishes three ways: as the site you may be reading, as a bound PDF with a cover and a contents page, and as a Word file with real styles. Every figure below is drawn from the reference pages, so a correction there reaches this page too.

> **How this handbook is built**
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> The site, the PDF and the Word file come from one `cudoc-export build` run and one set of design tokens. The configuration is a single file, `showcase.config.mjs`, beside the documents, and the summary table further down is assembled from the endpoint reference at build time.

What the API provides

Every station reports temperature, wind, humidity, pressure and precipitation on a fixed schedule, and the forecast model runs once an hour for every station. The API exposes that data as four data sets, each behind one or two endpoints.

Data set	Cadence	Retention	Endpoints
Current conditions	Every 10 minutes	Latest reading only	Current conditions
Forecasts	Hourly model run	48 hours and 7 days ahead	Hourly forecast, Daily forecast
Observation history	Every 10 minutes	Depends on the plan	Observation history
Station metadata	On change	Indefinite	Station directory, Station detail

Forecast horizons and history retention depend on the plan; see Plans compared.

Endpoints at a glance

Each row is one section of the endpoint reference. The method and the path are read from the table at the top of that section, and the description is its first paragraph, so this table cannot drift from the reference.

Endpoint	Method	Path	What it returns
Current conditions	GET	/v1/weather/current	Returns the latest observation from one station.

Endpoint	Method	Path	What it returns
Hourly forecast	GET	/v1/forecast/hourly	Returns up to 48 hourly forecast points for one station.
Daily forecast	GET	/v1/forecast/daily	Returns one forecast summary per calendar day, up to seven days ahead.
Observation history	GET	/v1/weather/history	Returns past observations from one station in a time range, oldest first.
Station directory	GET	/v1/stations	Lists the stations a key may read, with their location and status.
Station detail	GET	/v1/stations/{code}	Returns one station's metadata, including its sensors and time zone.
Weather alerts	GET	/v1/alerts	Lists active severe-weather alerts for a region, relayed from the national service.

Concepts

Stations and station codes

A station is one physical site with a fixed set of sensors. Every station has a code of the form `SITE-NN`: a three-letter site code and a two-digit sequence number, so a site with several masts has `BGO-01`, `BGO-02` and so on. Codes never change and are never reused; a station that stops reporting for good is marked `retired` and keeps its code.

Code	Site	Elevation	Status
`OSL-01`	Oslo	94 m	`active`
`BGO-01`	Bergen	12 m	`active`
`BGO-02`	Bergen	320 m	`maintenance`
`TRD-01`	Trondheim	8 m	`active`
`TOS-01`	Tromsø	100 m	`active`
`SVG-01`	Stavanger	4 m	`retired`

The list above is illustrative; the station directory is the source of truth for the stations a key may read.

Observations and forecasts

An observation is a reading a station actually made, stamped with the instant it was made. A forecast point is the model's expectation for a future instant, stamped with that instant and with the model run that produced it. The two are never mixed in one response: the current-conditions and history endpoints return observations, and the forecast endpoints return forecast points.

Units

Every numeric field is returned in metric units unless the request asks for `units=imperial`. Field names do not change with the unit system; only the values and the `units` object in the response do, so a client can always read the unit next to the value instead of assuming it.

Field	Metric	Imperial
`temperature`	°C	°F
`windSpeed`	m/s	mph
`pressure`	hPa	inHg
`precipitation`	mm	in
`visibility`	km	mi

Timestamps

Every instant is ISO 8601 in UTC with a `Z` suffix: `observedAt` on an observation, `validAt` and `issuedAt` on a forecast, `lastObservedAt` on a station. Converting to local time is the client's job; a station's IANA time zone is in its detail record so the conversion can be exact.

Versioning

The path prefix `/v1` names the version. Additive changes, such as a new field, a new optional parameter or a new endpoint, ship within `/v1` without notice, so a client must ignore fields it does not know. Any change to an existing shape is announced in the release notes at least ninety days before it takes effect, and the previous behaviour stays available for that period.

Conventions

The handbook uses a small set of marks, and each means the same thing in the site, the PDF and the Word file.

Mark	Meaning
A `New`, `Beta` or `Latest` badge after a heading	The section was added in the latest release, is in beta and may change on thirty days' notice, or describes the newest release.
A **Note** or **Tip** box	Background or a shortcut; skipping it costs nothing.
An **Important**, **Warning** or **Caution** box	Something that costs a key, a quota or data if ignored.
A **Ready** box	A checkpoint listing what a client needs before it moves on.
`code` in running text	A literal parameter, field, header or value.

Request and response samples are given as `curl`, Node.js and Python, and every JSON sample is a real response for the station `OSL-01`.

Where to start

1. Read Getting started for a key, a first request and a polling schedule.
2. Keep Rate limits open while you design that schedule; the request costs table tells you what each call weighs.
3. Handle every code in Error codes before you switch to a live key.
4. Check the release notes before upgrading a client.

> **Reading this on paper?**
> External links print their address after the text, and links between documents jump inside the bound file. The contents page at the front lists the page each document starts on.